AI-POWERED “INVISIBLE RESOURCE LOSS” DETECTOR MAKING HIDDEN WASTE VISIBLE FOR THE FIRST TIME

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PROBLEM STATEMENT

Urban areas lose a significant amount of water every day due to invisible losses such as underground leaks, pipe bursts, pressure anomalies, and inefficient distribution systems. These losses are often detected only after severe damage, leading to water scarcity, infrastructure failure, increased operational costs, and delayed response by authorities.

Existing monitoring systems rely on manual inspections or threshold-based alerts, which are reactive and ineffective for real-time prevention.

There is a critical need for an AI-driven system that can continuously monitor water usage data, detect abnormal patterns early, assess risk levels, and visually highlight high-risk zones to enable proactive intervention.

DETAILED SOLUTION DESCRIPTION

Invisible Loss AI is a real-time, AI-powered water loss detection dashboard designed for smart cities.

How the system works:

- Live Data Simulation - Water usage and pressure data from multiple zones are continuously updated to simulate real-time sensor feeds
- AI-Based Anomaly Detection - Machine learning algorithms analyze historical and live data to detect abnormal water consumption and pressure deviations
- Risk Scoring & Classification - Each zone is assigned a risk score and categorized as Low Risk, Medium Risk, High Risk,
- Interactive Visual Dashboard - Colored Risk Map highlighting zones based on risk severity

  - Time-series charts showing actual vs expected water usage

  - Donut, bar, and scatter charts for analytics and insights

  -AI-generated alerts explaining why a zone is risky
- Decision Support - Authorities can : Identify critical zones instantly, Monitor trends over time, Take preventive actions before failures occur

AI ELEMENTS & TOOLS USED

AI / ML Components:

- Anomaly Detection Logic
- Detects unusual water usage patterns
- Uses rolling averages and deviation analysis
- Risk Scoring Model
- Combines usage deviation and pressure anomalies
- Produces interpretable risk levels (Low / Medium / High)
- Explainable AI
- Generates human-readable explanations for each alert🛠Tools &

Tools & Technologies:

- Python
- Streamlit – interactive dashboard
- Pandas & NumPy – data processing
- Scikit-learn – anomaly detection logic
- Plotly – premium charts (line, bar, donut, scatter)
- PyDeck – interactive geospatial risk map
- GitHub – version control & deployment
- Streamlit Cloud – live prototype hosting

GitHub Link: https://github.com/chalini99/Water-loss

Demo Link: https:https://waterloss.streamlit.app/

https://waterloss.streamlit.app

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Invisible Water Loss AI

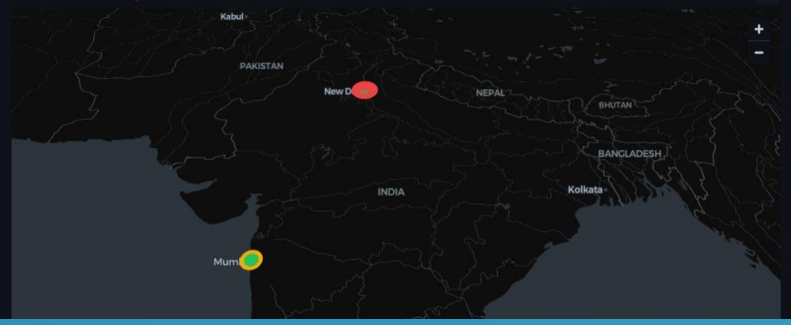
Real-time AI-powered monitoring of hidden water loss

High Risk Zones
3

Medium Risk Zones
1

Low Risk Zones
8

Live Water Risk Map



AI Insights

CRITICAL – Zone Z1
A sharp increase in water usage combined with reduced pressure was detected. This pattern strongly suggests a possible underground leak.

CRITICAL – Zone Z3
A sharp increase in water usage combined with reduced pressure was detected. This pattern strongly suggests a possible underground leak.

Manage app

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WARNING – Zone Z2
Abnormal water usage detected. Monitor closely.

Real-Time Water Loss Detection

Select Zone

Z1

Water Loss Analytics & Prediction

Preventive Actions (Water)

Recommended Action

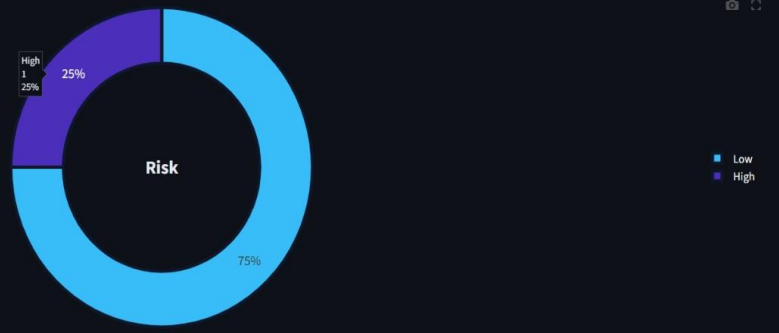
Inspect underground pipelines immediately

Preventive Actions (Water)

Recommended Action

Inspect underground pipelines immediately

Water Risk Distribution



Manage app

Daily Water Loss Severity
Risk Score
80
60
40
20
0
12:00
Dec 31, 2023
00:00
Jan 1, 2024
12:00
00:00
Jan 2, 2024
12:00
00:00
Jan 3, 2024
12:00
00:00
Jan 4, 2024
Date
Usage vs Pressure Correlation
Water Usage (Liters)
1800
1600
1400
1200
2.2
2.4
2.6
2.8
3
3.2
Pressure
80
60
40
20
0
Manage app

Severity Distribution
Low
3 (75%)
Medium
0 (0%)
High
1 (25%)
Critical
0 (0%)
Event Timeline
Events
1
0.8
0.6
0.4
0.2
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day
Total Events
Anomalies

THANK YOU